



Lecture 12: Weak law of large numbers

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1. Review
2. Markov, Chebyshev and Chernoff inequalities
3. Weak law of large numbers
4. Convergence in probability
5. Recitation
6. Summary



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- Sums of independent RVs

$$M_Z(s) = M_{X_1}(s) \cdots M_{X_n}(s)$$

- Sum of a *random* number of independent RVs

$$\mathbf{E}[Y] = \mathbf{E}[N] \mathbf{E}[X]$$

$$\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)$$

$$M_Y(s) = M_N(\log M_X(s)) = \sum_{n=0}^{\infty} (M_X(s))^n p_N(n)$$

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Theorem: Markov Inequality

If a RV X can only take nonnegative values, then

$$\mathbf{P}(X \geq a) \leq \frac{\mathbf{E}[X]}{a}, \quad \text{for all } a > 0.$$

- **Intuition:** If a *nonnegative* RV has a small mean, then the probability that it takes a large value must be small.

Theorem: Markov Inequality

If a RV X can only take nonnegative values, then

$$\mathbf{P}(X \geq a) \leq \frac{\mathbf{E}[X]}{a}, \quad \text{for all } a > 0.$$

- **Intuition:** If a *nonnegative* RV has a small mean, then the probability that it takes a large value must be small.
- **Proof:** Fix a positive number a ,

$$\begin{aligned}\mathbf{E}[X] &= \int_0^a x f_X(x) dx + \int_a^\infty x f_X(x) dx \\ &\geq 0 + a \int_a^\infty f_X(x) dx \\ &= a \mathbf{P}(X \geq a)\end{aligned}$$

- Let X be uniformly distributed in the interval $[0, 4]$ and note that $\mathbf{E}[X] = 2$. Then,

$$\mathbf{P}(X \geq 2) \leq \frac{2}{2} = 1, \quad \mathbf{P}(X \geq 3) \leq \frac{2}{3}, \quad \mathbf{P}(X \geq 4) \leq \frac{2}{4} = 0.5.$$

- By comparing with the exact probabilities,

$$\mathbf{P}(X \geq 2) = 0.5, \quad \mathbf{P}(X \geq 3) = 0.25, \quad \mathbf{P}(X \geq 4) = 0.$$

- The Markov inequality can be quite loose.

Theorem: Chebyshev's inequality

If a RV X has a finite mean μ and variance σ^2 , then

$$\mathbf{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

- **Intuition:** If a RV has small variance, then the probability that it takes a value far from its mean is also small.
- **Special case:** $\mathbf{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

Theorem: Chebyshev's inequality

If a RV X has a finite mean μ and variance σ^2 , then

$$\mathbf{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

- **Proof:** Re-write the definition of variance

$$\begin{aligned}\sigma^2 &= \int (x - \mu)^2 f_X(x) dx \\ &\geq \int_{-\infty}^{\mu-c} (x - \mu)^2 f_X(x) dx + \int_{\mu+c}^{\infty} (x - \mu)^2 f_X(x) dx \\ &\geq \int_{-\infty}^{\mu-c} c^2 f_X(x) dx + \int_{\mu+c}^{\infty} c^2 f_X(x) dx \\ &= c^2 \cdot \mathbf{P}(|X - \mu| \geq c)\end{aligned}$$

- When X is known to take values in a range $[a, b]$, we claim that $\sigma^2 \leq (b - a)^2/4$.
- Note that for any constant γ , we have

$$\mathbf{E}[(X - \gamma)^2] = \mathbf{E}[X^2] - 2\mathbf{E}[X]\gamma + \gamma^2.$$

- The above quadratic is minimized when $\gamma = \mathbf{E}[X]$. It follows that

$$\sigma^2 = \mathbf{E}[(X - \mathbf{E}[X])^2] \leq \mathbf{E}[(X - \gamma)^2], \quad \text{for all } \gamma.$$

- By letting $\gamma = (a + b)/2$, we obtain

$$\sigma^2 \leq \mathbf{E}\left[\left(X - \frac{a+b}{2}\right)^2\right] = \mathbf{E}[(X-a)(X-b)] + \frac{(b-a)^2}{4} \leq \frac{(b-a)^2}{4}.$$

Theorem: Chernoff inequality

If a RV X has a transform function $M_X(s)$, then

$$\mathbf{P}(X \geq a) \leq e^{-\max_{s \geq 0} (sa - \ln M_X(s))}.$$



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- **Proof:** Given a and $s \geq 0$, re-write the transform function

$$\begin{aligned} M_X(s) &= \int_{-\infty}^a e^{sx} f_X(x) dx + \int_a^{\infty} e^{sx} f_X(x) dx \\ &\geq 0 + e^{sa} \int_a^{\infty} f_X(x) dx \\ &= e^{sa} \mathbf{P}(X \geq a) \end{aligned}$$

from which we obtain

$$\mathbf{P}(X \geq a) \leq e^{-sa} M_X(s).$$

- Similarly, given a and $s \leq 0$,

$$\begin{aligned} M_X(s) &= \int_{-\infty}^a e^{sx} f_X(x) dx + \int_a^{\infty} e^{sx} f_X(x) dx \\ &\geq e^{sa} \int_{-\infty}^a f_X(x) dx + 0 \\ &= e^{sa} \mathbf{P}(X \leq a) \end{aligned}$$

from which we obtain

$$\mathbf{P}(X \leq a) \leq e^{-sa} M_X(s).$$

- The inequality

$$\begin{aligned}\mathbf{P}(X \geq a) &\leq \min_{s \geq 0} e^{-sa} M_X(s) \\ &= \min_{s \geq 0} e^{-(sa - \ln M_X(s))} \\ &= e^{-\max_{s \geq 0} (sa - \ln M_X(s))}\end{aligned}$$

- Denote $\phi(a) = \max_{s \geq 0} (sa - \ln M_X(s))$. If $a > \mathbf{E}[X]$, then $\phi(a) > 0$.

- The inequality

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- Denote $\phi(a) = \max_{s \geq 0} (sa - \ln M_X(s))$. If $a > \mathbf{E}[X]$, then $\phi(a) > 0$.

- **Proof:**

$$(1) \left. \phi(a) \right|_{s=0} = 0 - \ln 1 = 0;$$

$$(2) \left. \frac{d}{ds} (sa - \ln M_X(s)) \right|_{s=0} = a - \frac{1}{M_X(s)} \frac{d}{ds} M_X(s) \Big|_{s=0} = a - \mathbf{E}[X] > 0$$

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- Let $X_1, X_2, \dots$ be independent identically distributed (i.i.d.) RVs with finite mean μ and variance σ^2

$$M_n = \frac{X_1 + \dots + X_n}{n}$$

- $\mathbf{E}[M_n] =$
- $\text{var}(M_n) =$



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- $\mathbf{E}[M_n] = \mu$
- $\text{var}(M_n) = \frac{\sigma^2}{n}$



- Let $X_1, X_2, \dots$ be independent identically distributed (i.i.d.) RVs with finite mean μ and variance σ^2

$$M_n = \frac{X_1 + \dots + X_n}{n}$$

- $\mathbf{E}[M_n] = \mu$
- $\text{var}(M_n) = \frac{\sigma^2}{n}$
- We apply the **Chebyshev inequality** and obtain

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- For large n , the bulk of the distribution of M_n is concentrated near μ .

Theorem: The Weak Law of Large Numbers

Let $X_1, X_2, \dots$ be i.i.d. RVs with mean μ and variance σ^2 . For every $\epsilon > 0$, we have

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) = \mathbf{P}\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \epsilon\right) \rightarrow 0, \quad \text{as } n \rightarrow \infty$$

- **Intuition:** The sample mean of a large number of i.i.d. RVs is *very close* to the true mean, with high probability.

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- **Intuition:** The sample mean of a large number of i.i.d. RVs is *very close* to the true mean, with high probability.
- $X_1, X_2, \dots$ i.i.d. with finite mean μ and variance σ^2

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- M_n converges **in probability** to μ .

- f : the fraction of voters who support a particular officer
- i th (randomly selected) person polled:

$$X_i = \begin{cases} 1, & \text{if yes,} \\ 0, & \text{if no.} \end{cases}$$

- $M_n = (X_1 + \cdots + X_n)/n$, the fraction of “yes” in our sample
- Goal: 95% confidence of $\leq 1\%$ error

$$\mathbf{P}(|M_n - f| \geq .01) \leq .05$$

- Use Chebyshev's inequality:

$$\begin{aligned}\mathbf{P}(|M_n - f| \geq .01) &\leq \frac{\sigma_{M_n}^2}{(.01)^2} \\ &= \frac{\sigma_x^2}{n(.01)^2} \leq \frac{1}{4n(.01)^2}\end{aligned}$$

- If $n = 50,000$, then $\mathbf{P}(|M_n - f| \geq .01) \leq .05$ (conservative)

- $X_1, \dots, X_n$ i.i.d. finite variance σ^2
- Look at three variants of their sum:
 - $S_n = X_1 + \dots + X_n$: variance $n\sigma^2$
 - $M_n = \frac{S_n}{n}$: variance σ^2/n , converges “in probability” to $\mathbf{E}[X]$ (WLLN)
 - $\frac{S_n}{\sqrt{n}}$: constant variance σ^2 . Asymptotic shape?

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Theorem: Convergence of a Deterministic Sequence

Let $\{a_n\}$ (or $a_1, a_2, \dots$) be a deterministic sequence and a be another real number. We say that a_n converges to a , i.e., $\lim_{n \rightarrow \infty} a_n = a$, if for every $\epsilon > 0$, there exists n_0 , such that for every $n \geq n_0$, we have

$$|a_n - a| \leq \epsilon.$$

- **Intuition:**

“ a_n eventually gets and stays (arbitrarily) close to a ”

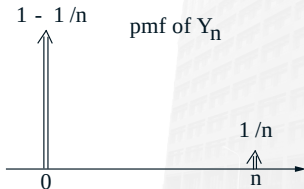
- **Question:** What is convergence “in probability” like?

Theorem: Convergence in Probability

Let Y_n (or $Y_1, Y_2, \dots$) be a sequence of RVs (not necessarily independent), and let a be a real number. We say that the sequence Y_n **converges to a in probability**, if for every $\epsilon > 0$, we have

$$\lim_{n \rightarrow \infty} \mathbf{P}(|Y_n - a| \geq \epsilon) = 0$$

- **Intuition:** Converges in probability to a number a : “(almost all) of the PMF/PDF of Y_n , eventually gets concentrated (arbitrarily) close to a ”



- Consider a sequence of independent RVs X_n that are uniformly distributed in the interval $[0, 1]$, and let

$$Y_n = \min\{X_1, \dots, X_n\} .$$

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- The sequence of values of Y_n cannot increase as n increases. Indeed, for $\epsilon > 0$, we have

$$\begin{aligned} \mathbf{P}(|Y_n - 0| > \epsilon) &= \mathbf{P}(X_1 \geq \epsilon, \dots, X_n \geq \epsilon) \\ &= \mathbf{P}(X_1 \geq \epsilon) \cdots \mathbf{P}(X_n \geq \epsilon) \\ &= (1 - \epsilon)^n . \end{aligned}$$

- In particular,

$$\lim_{n \rightarrow \infty} \mathbf{P}(|Y_n - 0| \geq \epsilon) = \lim_{n \rightarrow \infty} (1 - \epsilon)^n = 0 .$$

- Y_n converges to zero in probability.

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1. RV X is uniformly distributed between -1.0 and 1.0 . Let $X_1, X_2, \dots$ be independent identically distributed RVs with the same distribution as X . Determine which, if any, of the following sequences (all with $i = 1, 2, \dots$) are convergent in probability.

(a) $U_i = \frac{X_1 + X_2 + \dots + X_i}{i}$

(b) $W_i = \max\{X_1, \dots, X_i\}$

(c) $V_i = X_1 \cdot X_2 \cdot \dots \cdot X_i$

2. Demonstrate that the Chebyshev inequality is tight, that is, for every $\mu, \sigma > 0$, and $c \geq \sigma$, construct a RV X with mean μ and standard deviation σ such that

$$\mathbf{P}(|X - \mu| \geq c) = \frac{\sigma^2}{c^2} .$$

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- Chebyshev's inequality

$$\mathbf{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- Weak law of large numbers

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- Convergence “in probability”

$$\lim_{n \rightarrow \infty} \mathbf{P}(|Y_n - a| \geq \epsilon) = 0$$

- Readings: Section 5.1-5.3
- Assignment: Problem Set 12
- Next lecture: Central limit theorem, strong law of large numbers